

DecisionIQ Master Handbook

Enterprise Operational Intelligence Platform

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CHAPTER 1: Introduction to DecisionIQ

1.1 The Monday Morning Crisis: A Story of Blind Flight

It is Monday morning, 8:00 AM. Arthur Pendelton, the CEO of TechNova Global - a mid-market enterprise supplying cloud infrastructure and hardware servers - sits at his mahogany desk. He opens his executive dashboard, expecting to see the steady upward climb of Q2 revenue. Instead, he is met with a flashing crimson indicator: Net Sales Revenue has declined by 8.5% over the past six weeks.

Arthur's heart sinks. He immediately calls an emergency conference with his department directors:

- **The VP of Sales** is defensive: "Our pipeline is healthy, and the sales team is closing deals. If revenue is down, it's not because we aren't selling."
- **The VP of Finance** points to a 20% spike in refunds.
- **The VP of Customer Success** warns of a ticket storm: CSAT has plummeted to 1.8 out of 5.0, and clients are threatening subscription cancellations.
- **The Director of Logistics** explains that the West Coast Hub is understaffed and Apex Technology (supplier) is weeks behind schedule.

The meeting dissolves into arguments. Sales is looking at Salesforce; Finance is looking at NetSuite; Logistics is tracking shipping portals; CS is in Zendesk. Arthur realizes TechNova is in blind flight. The data exists, but it is siloed. No one can trace how a delay at a supplier's warehouse in Asia cascades into a refund request in California and a customer churn risk in New York.

Arthur doesn't need another dashboard; he needs a central intelligence layer. He needs DecisionIQ.

1.2 The Data Silo & Dashboard Fallacy

A Data Silo is an isolated repository of data that is accessible by one department but cut off from the rest of the organization. Silos are the natural byproduct of specialized software.

Traditional dashboards fail because they are:

1. **Descriptive, Not Diagnostic:** They tell you *what* happened, but not *why*.
2. **No Actionable Guidance:** They display stats but leave decisions to human guesswork.
3. **Disconnected from Future Risk:** They look backward rather than predicting customer churn or inventory stockouts.

1.3 Decision Intelligence Defined

Decision Intelligence (DI) is a practical discipline that improves decision-making by using data engineering, advanced SQL, machine learning, and business logic.

Feature	Traditional Dashboards	Business Intelligence (BI)	Decision Intelligence (DI)
Primary Goal	Display operational data streams	Summarize historical trends	Optimize business outcomes
Output Type	Raw numbers and simple charts	Aggregated KPIs and trends	Root-cause analysis & actions
Temporal Focus	Present (Real-time snapshots)	Past (Historical reporting)	Future & Prescriptive (Forecasts)
User Action	Manual inspection and guessing	Manual drill-downs & audits	Selecting prescribed actions

1.4 TechNova Case Study: The Q2 2025 Chain Reaction

During Q2 2025, TechNova experienced a supply chain breakdown:

- Supplier Delay:** Supplier Apex Technology Corp reliability drops from 95% to 72%, delaying parts by 12 days.
- Warehouse Bottleneck:** West Coast Center WH002 safety stock runs low, and understaffing causes shipping delays.
- Fulfillment Failure:** Delivery status for West Coast shipments becomes "Late" for 55% of orders.
- Ticket Storm:** Customers complain about logistics delays. Support CSAT rating crashes to 1.8/5.0.
- Financial Leakage:** Customers cancel orders, leading to an 18% refund spike and \$245,000 lost in region REG01.

1.5 Objectives & Technologies

The DecisionIQ codebase is designed around:

- Data Modeling:** 14-table relational schema (regions, employees, customers, products, warehouses, suppliers, inventory, marketing, orders, details, payments, shipments, support, metrics).
- ETL & Quality:** Python validator enforcing PK, FK, and range checks, and mapping duplicate customer accounts.
- Predictive ML:** Churn probability models, weekly revenue forecasters, and supplier delay risk classifiers.
- Visualization:** Streamlit dashboard built on McKinsey/BCG design tokens.

CHAPTER 2: Data Architecture & Database Design

2.1 The Data Warehouse Schema Design

DecisionIQ uses a normalized Snowflake schema to map the entire value chain. This design structures logical lookup tables (dimensions) and transactional tables (facts) to avoid redundancy and enforce data integrity.

[illegible]

- ****Parent Lookup Tables:****
- ``regions``: ``region_id`` (PK), ``region_name``, ``country``, ``manager_name``.
- ``products``: ``product_id`` (PK), ``product_name``, ``sku``, ``category``, ``unit_price``, ``unit_cost``, ``status``. Enforces ``CHECK (unit_price >= 0 AND unit_cost >= 0)``.
 - ``suppliers``: ``supplier_id`` (PK), ``supplier_name``, ``lead_time_days``, ``reliability_score``. Enforces ``CHECK (reliability_score >= 0.0 AND reliability_score <= 1.0)``.
- ****Child Transactional Tables:****
- ``customers``: ``customer_id`` (PK), ``company_name``, ``contact_email`` (UNIQUE), ``customer_segment`` (SMB, Enterprise, Strategic), ``region_id`` (FK referencing regions).
- ``orders``: ``order_id`` (PK), ``customer_id`` (FK), ``order_date``, ``status`` (Completed, Shipped, Processing, Cancelled, Refunded), ``total_amount`` (CHECK >= 0).
- ``shipments``: ``shipment_id`` (PK), ``order_id`` (FK), ``warehouse_id`` (FK), ``supplier_id`` (FK), ``delivery_status`` (On Time, Late, In Transit, Returned).

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To query millions of rows instantly, we configure indexes on frequently joined columns:

```
```sql
CREATE INDEX idx_orders_customer_id ON orders(customer_id);
CREATE INDEX idx_orders_order_date ON orders(order_date);
CREATE INDEX idx_shipments_actual_delivery ON shipments(actual_delivery_date);
```
```

Indexes decrease query execution times from $O(N)$ table scans to $O(\log N)$ B-Tree lookups, ensuring C-suite dashboards render in milliseconds.

CHAPTER 3: Ingestion, Validation, and ETL Pipelines

3.1 The Ingestion Architecture

The ETL (Extract, Transform, Load) pipeline extracts raw operational CSVs, runs validation checks, cleans anomalies, and bulk-loads data.

```
```
[Raw CSVs] --> [Deduplication & Mapping] --> [Range & FK Checks] --> [SQLAlchemy Ingest]
```
```

3.2 Master Data Management (MDM): Account Consolidation

A common data issue is duplicate customer registrations. For example, John Doe registers as `CUST002` and `CUST045` under the same email.

Simply deleting the duplicate record `CUST045` results in a foreign key violation in the `orders` table because existing orders reference the deleted ID.

Our ETL solves this by building an ID mapping dict and re-keying child rows:

```
```python
is_duplicate = customers_raw.duplicated(subset=["contact_email"], keep="first")
master_ids = customers_raw[~is_duplicate][["contact_email", "customer_id"]].rename(columns={"customer_id":
"master_customer_id"})
duplicate_ids = customers_raw[is_duplicate][["contact_email", "customer_id"]]
mapping = dict(zip(duplicate_ids.merge(master_ids, on="contact_email")["customer_id"],
mapping_df["master_customer_id"]))

Re-key orders and support tickets before deleting duplicates
orders["customer_id"] = orders["customer_id"].replace(mapping)
support["customer_id"] = support["customer_id"].replace(mapping)
```
```

3.3 Programmatic Quality Checks

The `validator.py` script runs three core audits before database entry:

1. **Null Audits:** Enforces non-null primary keys.
2. **Range Audits:** Ensures prices and quantities are strictly positive.
3. **Referential Integrity Audits:** Ensures every `customer_id` in the `orders` table exists in the master `customers` dimension table.

CHAPTER 4: Advanced SQL & Predictive Machine Learning

4.1 Advanced Analytics in SQL

We run cohort and segmentation calculations directly in the database for maximum efficiency:

- **Customer Cohort Retention:** Computes monthly retention curves, tracing what percentage of sign-up cohorts continue purchasing up to 12 months later.
- **RFM Quintiles:** Uses `NTILE(5)` over recency, frequency, and monetary metrics to tag customer accounts:

$$\text{RFM Score} = \text{NTILE}_5(\text{Recency}) + \text{NTILE}_5(\text{Frequency}) + \text{NTILE}_5(\text{Monetary})$$
- **Recursive Org Hierarchy:** Walks down `reports_to` trees to compute rollup revenue generated by subordinates under each regional manager.

4.2 Machine Learning Suite

- **Customer Churn Classifier:** Trains a Random Forest model on RFM variables and support ticket history to calculate active customer churn probabilities.
- **Revenue Time-Series Forecaster:** Ridge Regression utilizing trend, month-of-year seasonality, weekly revenue lags (\$t-1\$ to \$t-4\$), and 4-week rolling averages to forecast the next 12 weeks of enterprise sales.
- **Supplier Risk Classifier:** Assesses the delay probability of outstanding component orders using carrier, supplier history, and destination warehouse features.

CHAPTER 5: Insight Engine & C-Suite Visualization

5.1 The Automated Diagnostic Engine

The Core of DecisionIQ is the `InsightEngine`. When Net Revenue drops, the engine queries the database, correlates operational anomalies, and generates structured executive reports.

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Insight Diagnostic Pipeline

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| Revenue Drop MoM | --> | Late Shipments Spike | --> | Supplier delay (Apex) |
|---------------------|-----|----------------------|-----|-----------------------|
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| Action: Dual Source | <-- | CSAT Crashed (1.8) | <-- | CS ticket spike |
| Est value: \$180k | | Churn risk in SMB | | Category: Logistics |
| | | | | |
| | | | | |

The output contains:

- **Anomalous Event:** Q2 2025 revenue declined by 8.5%.
- **Root Cause:** Apex Technology Corp delays caused a 55% late shipment rate at warehouse WH002, triggering logistics ticket storms and a CSAT crash to 1.8.
- **Financial Leakage:** Calculated at \$245,000 in refunds.
- **Prescriptive Recommendations:** Dual-source components from EuroChip AG (SUP004), reallocate 15% safety stock to WH002, and trigger proactive retention campaigns.

5.2 Executive Consulting Dashboard Design

The dashboard is styled using professional consulting standards:

- **Aesthetic Palette:** Primary Navy Blue (`#1B2A4A`), Accent Gold (`#C5A059`), Cool Slate (`#F4F5F7`), and Soft Crimson (`#C62828`).
- **Layout Framework:** Avoids colorful student grids. It uses cards with rounded corners, ample white space, and clear headers focusing on answering business questions.
- **Interactive Controls:** Supports regional filters, cohort heatmaps, and ML-predicted high-churn risk lists for immediate account outreach.